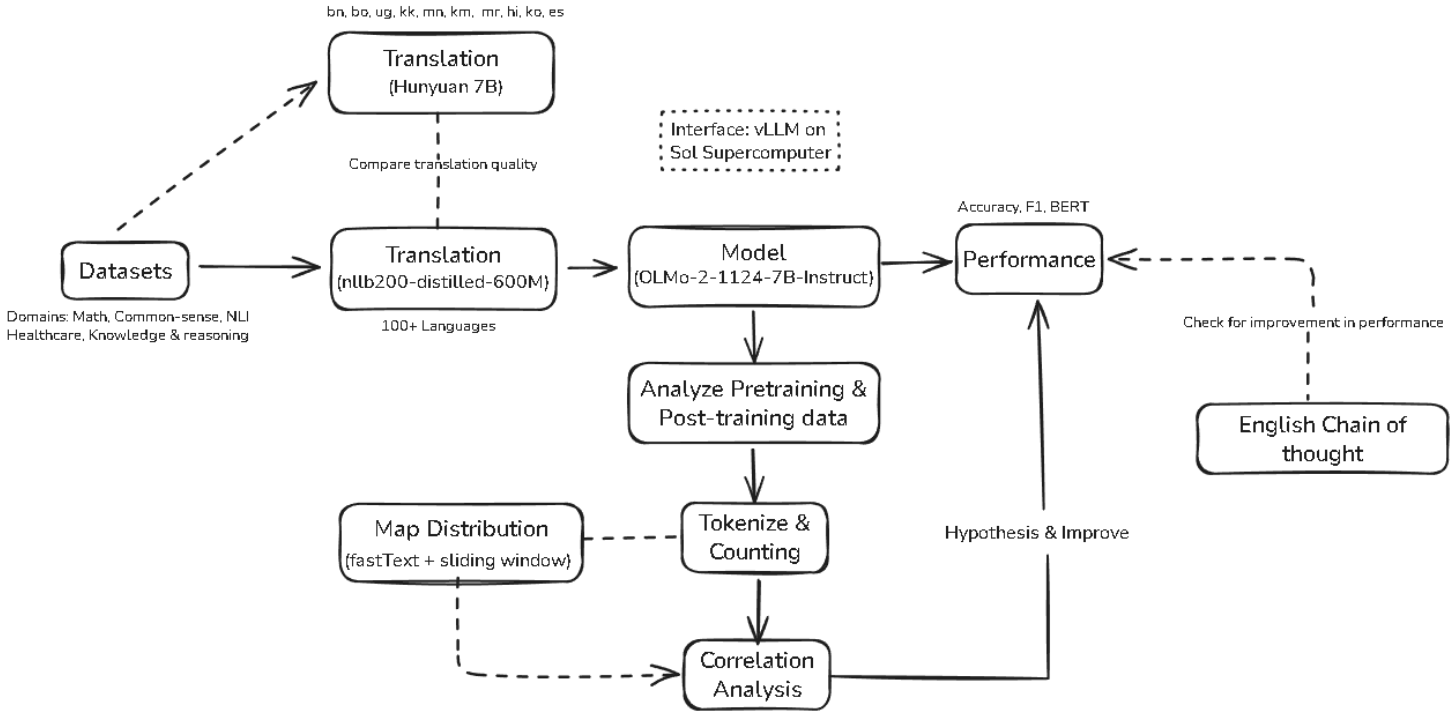
Identifying LLM Performance Gaps Across Languages

1. Problem statement

Large language models (LLMs) trained primarily on English data often demonstrate significantly stronger performance in English compared to other languages. This project investigates the extent and underlying reasons for this performance gap across a range of domains. The key objectives are to (a) establish reliable baselines to measure and compare performance across languages and tasks, (b) quantify the correlation between performance and language-specific pre-training factors, specifically analyzing the impact of severe data asymmetry; and (c) empirically distinguish whether performance failures stem from linguistic incomprehension or a foundational lack of reasoning adaptation in non-English contexts.

In future phases, the goal is to extend this analysis to a wider range of languages and evaluate strategies for reducing multilingual disparities. By systematically examining how data distribution and tokenization affect reasoning, this project aims to demonstrate that fine-tuning alone is insufficient to correct architectural bias. Ultimately, these findings serve as a reference benchmark, providing the empirical evidence necessary to shift the paradigm of future model training toward a truly language-agnostic foundation.

2. Approach & Implementation Details



Dataset Construction and Task Coverage

We evaluated 10 datasets spanning five reasoning domains: Math (MGSM, SVAMP), Commonsense (XCOPA, XCSQA, XCODAH), Natural Language Inference (XNLI), Healthcare QA (HealthQA, LiveQA, MedicationQA), and General Knowledge & Reasoning (MKQA, MMLU, M-ARC). Phase-1 covered six languages; in Phase-2, we scaled to ~100 languages, translating the native datasets, enabling robust cross-lingual benchmarking.

100 languages considered: als, arz, ast, azb, ceb, ckb, ilo, lmo, mai, min, scn, vec, war, yue, af, am, as, ba, be, bn, bo, bs, bg, ca, cs, cy, da, de, el, eo, et, eu, fi, gd, ga, gl, gn, gu, hi, ht, he, hr, hu, hy, id, is, it, jv, ja, kn, ka, kk, km, ko, ky, lo, li, lt, lb, ml, mr, mk, mt, my, nl, nn, oc, pa, pl, pt, ro, ru, sa, si, sk, sl, sd, so, sc, sr, su, sv, ta, tt, te, tg, tl, th, tk, tr, ug, uk, ur, vi, yo, sw, yi, fr, es, zh

Translation Pipeline

Used NLLB-200-Distilled-600M model by Meta for multilingual translation. For even better translation quality, Hunyuan-MT-7B (WMT25 winner in 30/31 tracks) translated every sample 5 times along with the Hunyuan-MT-Chimera-7B ensemble applied on 10 languages to benchmark translation quality.

Evaluating Translation Accuracy

To ensure that any evaluation drop was due to the model and not translation drift, we conducted:

- Comparison between English and translated languages
    - COMET score (Adequacy), LABSE similarity
- NLLB-based back-translation, Hunyuan-based back-translation
    - BLEU Score, METEOR score, ROUGE-L, BERTScore

Model Evaluation

All experiments used allenai/OLMo-2-1124-7B-Instruct, with VLLM enabling optimized memory usage, flash attention, fast decoding, and continuous batching. To reduce variance, self-consistency inference was applied with 20 independent runs per sample at settings (temperature =0.7, top_k =20, n = 20), aggregating outputs via majority voting.

English Chain-of-Thought (CoT)

To test reasoning transfer, the model was prompted to reason in English while keeping input and output in the same language. This was compared against standard prompting, where both reasoning and responses occurred in the native language, allowing us to quantify improvements attributable to English-based reasoning exposure.

Token Exposure Analysis

To analyze structural multilingual disparities, per-language token exposure was measured in the pre-training and post-training datasets of the Olmo model.

- **Language Detection:** fastText with a 5-token sliding window.
- **Pre-training Sampling:** 0.1% global sampling of dolmino-mix-1124 (843B tokens).
- **Post-training:** Full dataset including TULU SFT, preference-mix, and RLVR-math.
- **Rank correlations** between token counts and downstream accuracy quantified the impact of pre-training exposure on multilingual performance.

Large-Scale Multilingual Evaluation

Combining dataset expansion, translation accuracy, self-consistent inference, CoT experiments, and token-exposure analysis, we executed a full evaluation across ~100 languages. This methodology enabled precise attribution of reasoning failures to model limitations versus translation noise, providing one of the most comprehensive multilingual benchmarks for analyzing LLM performance equity.

3. Results

Domain	Dataset	Evaluation metric	en	fr	es	zh	hi	ko
Math	MGSM	Accuracy	0.676	0.276	0.184	0.156	0.084	0.140
	SVAMP	Accuracy	0.599	0.461	0.535	0.377	0.208	0.206
NLI	XNLI	Accuracy	0.875	0.720	0.737	0.623	0.342	0.596
Common Sense	XCOPA	Accuracy	0.925	0.790	0.750	0.764	0.480	0.400
	XCSQA	Accuracy	0.617	0.448	0.457	0.364	0.285	0.240
		F1	0.587	0.418	0.429	0.332	0.228	0.211
	XCODAH	Accuracy	0.550	0.357	0.350	0.270	0.267	0.200
		F1	0.548	0.323	0.301	0.231	0.192	0.183
General Knowledge Reasoning	MKQA	Accuracy	0.440	0.395	0.415	0.380	0.360	0.340
		F1	0.470	0.425	0.445	0.405	0.385	0.365
	MMLU	Accuracy	0.613	0.252	0.260	0.206	0.197	0.184
	M_ARC	Accuracy	0.631	0.294	0.307	0.233	0.219	0.181
HealthCare	healthqa	BERT Score	0.635	0.625	0.615	0.610	0.585	0.524
	liveqa	BERT Score	0.640	0.630	0.620	0.623	0.608	0.545
	medicationqa	F1	0.480	0.300	0.320	0.230	0.180	0.160
		BERT Score	0.380	0.280	0.290	0.240	0.220	0.210

Table 1: Baseline Evaluation on 6 Languages

Domain	Dataset	Evaluation metric	en	fr	es	zh	hi	ko
Math	MGSM	Accuracy	-	0.364	0.196	0.196	0.076	0.136
	SVAMP	Accuracy	-	0.607	0.548	0.321	0.253	0.207
NLI	XNLI	Accuracy	-	0.731	0.739	0.628	0.487	0.502
Common Sense	XCOPA	Accuracy	-	0.637	0.589	0.510	0.308	0.320
	XCSQA	Accuracy	-	0.605	0.410	0.295	0.205	0.165
		F1	-	0.556	0.385	0.265	0.170	0.140
	XCODAH	Accuracy	-	0.620	0.400	0.310	0.215	0.185
		F1	-	0.590	0.370	0.280	0.200	0.140
General Knowledge Reasoning	MKQA	Accuracy	-	0.420	0.435	0.395	0.370	0.350
		F1	-	0.455	0.475	0.425	0.405	0.380
	MMLU	Accuracy	-	0.300	0.305	0.248	0.222	0.206
	M_ARC	Accuracy	-	0.320	0.352	0.288	0.243	0.194
HealthCare	healthqa	BERT Score	-	0.645	0.635	0.625	0.605	0.544
	liveqa	BERT Score	-	0.650	0.640	0.638	0.628	0.565
	medicationqa	F1	-	0.340	0.360	0.270	0.220	0.200
		BERT Score	-	0.370	0.380	0.330	0.310	0.300

Table 2: English CoT Evaluation on 6 Languages

Language	Token count	Token Distribution	Avg Perf
en	174624114	94.57%	64.10%
fr	1381649	0.75%	44.40%
es	1085026	0.59%	43.80%
zh	245423	0.13%	36.50%
hi	49483	0.03%	27.40%
ko	36619	0.02%	27.60%
Spearman Correlation - 0.942			

Table 3: Correlation of tokens in Pretraining Data (185M tokens) vs Performance for 6 languages

Language	Token count	Token Distribution	Avg Perf
en	727060314	65.90%	64.10%
fr	27108706	2.46%	44.40%
es	30873895	2.80%	43.80%
zh	39871061	3.61%	36.50%
hi	4192851	0.38%	27.40%
ko	9719077	0.88%	27.60%
Spearman Correlation - 0.771			

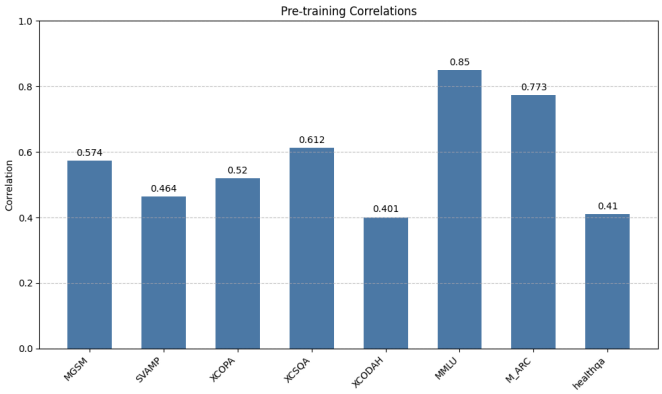
Table 4: Correlation of tokens in Post Training Data (1.1B tokens) vs Performance for 6 languages

language	token_count	token_pct	accuracy	rank_token	rank_accuracy
en	178,269,082.00	0.952238	0.8975	1	1
fr	1,406,111.00	0.007511	0.7225	2	2
de	1,302,288.00	0.006956	0.71	3	3
es	1,072,049.00	0.005726	0.685	4	6
it	703,479.00	0.003758	0.6525	5	10
ru	526,991.00	0.002815	0.6925	6	5
pt	382,306.00	0.002042	0.705	7	4
nl	226,390.00	0.001209	0.66	8	8
ca	206,862.00	0.001105	0.6425	9	11
pl	192,912.00	0.00103	0.5925	10	20
ceb	192,396.00	0.001028	0.565	11	26
ja	192,224.00	0.001027	0.6075	12	17
zh	184,040.00	0.000983	0.635	13	12
tr	183,908.00	0.000982	0.5525	14	30
sv	158,609.00	0.000847	0.62	15	14

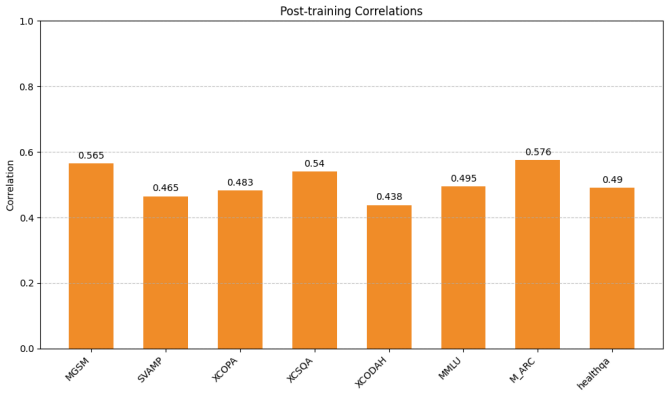
Table 5(a): Token count vs Accuracy (Pre-training) - XCOPA for 100 languages

lang	token_count	token_pct	accuracy	rank_token	rank_accuracy
en	729,015,568.00	0.679844	0.8975	1	1
ru	67,797,191.00	0.063224	0.6925	2	5
es	30,959,373.00	0.028871	0.685	3	6
zh	29,807,725.00	0.027797	0.635	4	12
fr	27,377,465.00	0.025531	0.7225	5	2
de	21,144,342.00	0.019718	0.71	6	3
it	19,934,681.00	0.01859	0.6525	7	10
tr	13,487,453.00	0.012578	0.5525	8	30
pt	12,171,325.00	0.01135	0.705	9	4
ja	11,038,205.00	0.010294	0.6075	10	17
ko	9,155,948.00	0.008538	0.62	11	14
uk	7,983,709.00	0.007445	0.5975	12	19
ceb	7,384,591.00	0.006887	0.565	13	26
vi	6,274,382.00	0.005851	0.63	14	13
ta	5,984,867.00	0.005581	0.4975	15	49

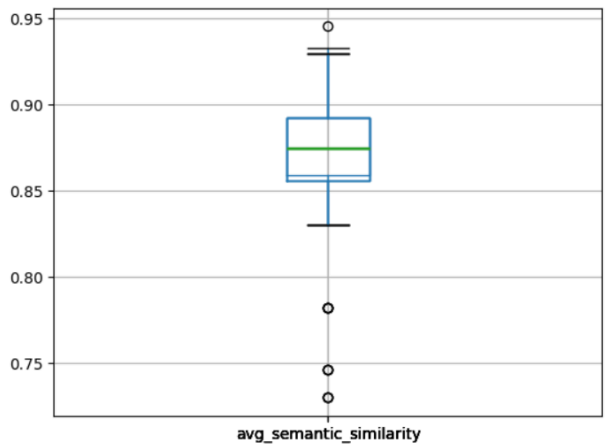
Table 5(b): Token count vs Accuracy (Post-training) - XCOPA for 100 languages



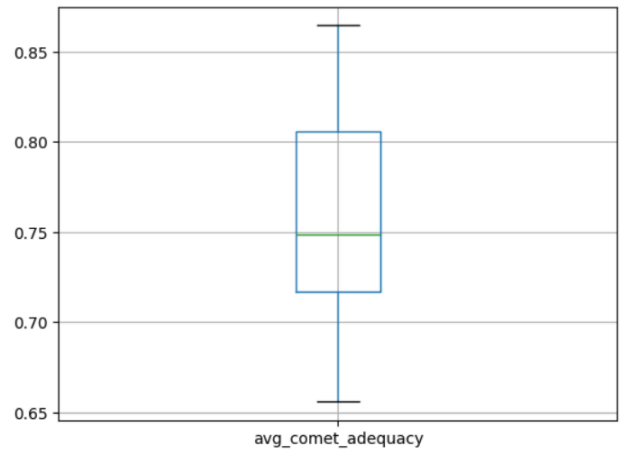
Graph 1: Rank Correlation of Token count in pre training data vs Performance for 100 languages



Graph 2: Rank Correlation of Token count in post training data vs Performance for 100 languages



Graph 3: MGSM direct translation accuracy: Average LaBSE-based semantic similarity computed across 100 languages.



Graph 4: MGSM direct translation accuracy: Average COMET adequacy score computed across 100 languages.

Language	MGSM		MMLU		XCOPA	
	Hunyuan	NLLB	Hunyuan	NLLB	Hunyuan	NLLB
bn	0.190	0.250	0.659	0.811	0.256	0.420
bo	0.020	0.080	0.337	0.307	0.267	0.184
ug	0.100	0.090	0.118	0.143	0.248	0.340
km	0.070	0.090	0.313	0.398	0.190	0.317
kk	0.120	0.170	0.660	0.805	0.205	0.343
mr	0.190	0.230	0.401	0.517	0.275	0.470
hi	0.130	0.390	0.780	0.936	0.301	0.534
ko	0.200	0.180	0.384	0.48	0.327	0.407
es	0.200	0.260	0.574	0.735	0.391	0.636

Table 6(a): BLEU scores for back-translation–based translation accuracy comparison between Hunyuan and NLLB.

Language	MGSM		MMLU		XCOPA	
	Hunyuan	NLLB	Hunyuan	NLLB	Hunyuan	NLLB
bn	0.590	0.580	0.410	0.418	0.619	0.679
bo	0.162	0.353	0.278	0.361	0.563	0.435
ug	0.445	0.379	0.503	0.488	0.572	0.594
km	0.411	0.382	0.359	0.377	0.528	0.594
kk	0.504	0.476	0.308	0.311	0.520	0.591
mr	0.601	0.569	0.405	0.437	0.628	0.724
hi	0.528	0.691	0.485	0.572	0.646	0.768
ko	0.604	0.530	0.374	0.352	0.684	0.660
es	0.639	0.572	0.416	0.441	0.700	0.820

Table 6(b): ROUGE-L scores for back-translation–based translation accuracy comparison between Hunyuan and NLLB.

4. Analysis of Results and Findings

Baseline Evaluation Across Domains

Across all evaluated domains (Math, Commonsense, NLI, Reasoning, and Healthcare)**,** English consistently achieves the strongest performance. *Table 1* clearly shows that for every dataset and metric (Accuracy, F1, BERTScore), English remains the highest-performing language. Languages such as **French, Spanish, and Chinese** form a second tier, while **Hindi and Korean** consistently fall behind, even after translation correction using **NLLB**. These findings confirm the existence of a systematic performance gap between English and other languages, despite identical prompts, identical datasets, and identical inference settings.

Across the full 100-language baseline evaluation, **the same trends observed in smaller subsets become clearer and more systematic**. **Accuracy drops steadily as languages diverge from English**, with high-resource Latin/Cyrillic languages forming the strongest cluster. For example, French, German, and Spanish consistently remain in the top tier, while structurally distant or low-resource languages such as Khmer, Burmese, and Tibetan fall sharply. Script effects are much more pronounced at this scale: Romance and Germanic languages remain tightly grouped at higher scores, whereas languages using Brahmic scripts (Tamil, Kannada, Sinhala) and logographic or non-segmenting scripts (Burmese, Khmer) show significantly reduced accuracy. Several outliers, such as Basque, Armenian, and Burmese, demonstrate disproportionately low performance relative to their resource level.

Chain-of-Thought in English vs Native Language

Table 2 shows the evaluation after applying **English Chain-of-Thought (CoT)** prompting. Under this setting, the model performs its intermediate reasoning in English with the rest of the settings intact. **Spanish, French, and Chinese** show noticeable improvements. This suggests that the model possesses stronger internal reasoning pathways in English, and leveraging them mitigates some multilingual weaknesses. **Hindi and Korean** do not benefit, and in several tasks, the performance remains nearly unchanged. The likely reasons could be due to (i) low token exposure in training data, (ii) script differences (Devanagari, Hangul), and (iii) weaker alignment between their embeddings and English embeddings. When extended to 100 languages, English CoT yields generally positive but uneven gains. High-resource languages like French, Spanish, and Chinese continue to benefit substantially from English reasoning, while mid-resource languages show mixed outcomes; those closer to English in syntax (Romance, Germanic) improve more consistently than agglutinative or morphologically complex languages such as Turkish, Finnish, or Hungarian. For many low-resource languages, gains are small or unstable, and in some cases, English CoT slightly hurts performance due to semantic drift. In summary, the majority of the languages show improvement, indicating that English CoT is broadly helpful but not universally effective. The full 100-language analysis confirms that CoT success is strongly tied to **language resource level and structural similarity to English**.

Correlation Between Token Exposure and Performance

The pre-training and post-training token-exposure analyses (*Table 3, 4,* and *Graphs 1, 2*) demonstrate a **very strong positive rank correlation** between token count and average performance: **Pre-training Spearman correlation:** 0.4 - 0.85**, Post-training Spearman correlation:** 0.49-0.7 This is one of the clearest findings in the report: **"Languages with more tokens in the model's training corpus exhibit substantially higher downstream accuracy"**. For languages like French and Spanish (*Table 5a, 5b*), the high token count and proximity to English and language script reflect the performance across the tasks. Conversely, Hindi and Korean, despite moderate real-world populations, have extremely small token exposure and suffer large accuracy drops. Across all 100 languages, token exposure remains one of the strongest predictors of accuracy. However, the full dataset shows that exposure alone is not sufficient; some high-token languages still underperform due to script or morphological complexity, while mid-resource languages like Indonesian or Swedish perform disproportionately well. Clear tiers emerge, from high-exposure European languages at the top to low-resource languages with unstable accuracy at the bottom.

Translation Quality Analysis (Hunyuan vs NLLB)

The translation-quality comparison includes BLEU, ROUGE-L, and METEOR scores for 10 languages. Significant patterns emerge:

BLEU and ROUGE-L: NLLB consistently achieves higher BLEU and ROUGE-L scores than Hunyuan.
Example (**bn**, from *Table 12*): BLEU: 0.25 (NLLB) vs 0.19 (Hunyuan), ROUGE-L: 0.5804 (NLLB) vs 0.5902 (Hunyuan - exception). To summarize, NLLB is more literal, which boosts n-gram-based metrics like BLEU and ROUGE-L.

METEOR: Hunyuan performs **substantially better**: **bn:** 0.5845 (Hunyuan) vs 0.4784 (NLLB), **hi:** 0.5185 (Hunyuan) vs 0.4619 (NLLB). This is expected because METEOR rewards: (i) synonym matching, (ii) stemming, and (iii) flexible word order. Hunyuan's generative structure allows paraphrasing, so METEOR favours it.

Hunyuan vs NLLB Across Downstream Tasks

Tables 6a and *6b* compare MGSM, MMLU, and XCOPA scores using Hunyuan vs NLLB translations. A unified pattern emerges: **For MGSM and MMLU, Hunyuan performs slightly better** (likely due to clearer phrasing). **For XCOPA, NLLB sometimes wins** (literal translations align better with entailment-style tasks). This supports that **task structure interacts with translation style**, literal translations benefit classification tasks, while generative reasoning tasks benefit more fluent phrasing.

Back-Translation Consistency

Charts (*Graph 3* and *Graph 4*) show: (i) Semantic similarity (LABSE) clusters tightly around ~0.88–0.92, (ii) COMET adequacy averages ~0.75–0.82**.** Languages like Spanish and French maintain high adequacy; low-resource or complex scripts like **Tibetan (bo)** and **Kazakh (kk)** show noticeably lower adequacy. This indicates that translation noise is *not* the primary cause of poor performance; it is consistent across most languages.

NLLB and Hunyuan show complementary strengths. NLLB is more literal and structurally faithful, leading to higher stability on strict n-gram metrics like BLEU and ROUGE-L, which favour word-level alignment. Hunyuan, by contrast, produces more fluent and semantically richer translations, reflected in higher METEOR scores that reward paraphrasing and synonymy. Its main drawback is that, as a causal LLM, it can occasionally generate empty, truncated, or partially incorrect outputs, so additional validation is needed. Overall, Hunyuan provides better semantic adequacy for the 34 languages it supports, while NLLB remains important for wide language coverage. For unsupported languages, we rely on models like Seed-X, X-ALMA, or SeamlessM4T-Large.

5. Contributions

Literature Review and Dataset selection (all domains)	Amrit, Bhavan, Karthik, Omkar
Experimental Design and Baseline Evaluation	Amrit (Math), Bhavan (Reasoning), Karthik (Common Sense), Omkar (Health)
Translation of Datasets to 100 languages	Amrit, Bhavan, Karthik, Omkar
English Chain of Thought Experiment and Analysis	Amrit, Bhavan, Karthik, Omkar
Per-language token distribution calc (Pre + Post training)	Amrit
Hypothesis formulation	Amrit, Bhavan, Karthik, Omkar
Self-Consistency Evaluation	Amrit (MGSM, SVAMP), Bhavan (MMLU, MARC), Karthik (XCOPA, XCSQA, XCODAH), Omkar (HealthcareQA)
NLLB & Hunyuan Back translation evaluation	Amrit, Bhavan, Karthik

6. References

1. Team OLMo et al. (2022). *OLMo 2 Furious*. arXiv preprint arXiv:2501.00656.

2. Zheng, M., et al. (2025). *Hunyuan-MT Technical Report*. arXiv:2509.05209.

3. NLLB Team, Costa-jussà, M. et al. (2022). *No Language Left Behind: Scaling Human-Centered Machine Translation*. arXiv:2207.04672.

4. X. Wang et al. (2022). *Self-Consistency Improves Chain of Thought Reasoning in Language Models*. arXiv preprint arXiv:2203.11171.

5. P. Bojanowski, E. Grave, A. Joulin, and T. Mikolov (2017). Enriching Word Vectors with Subword Information. Transactions of the Association for Computational Linguistics, vol. 5, pp. 135–146.

6. S. Singh et al. (2024). *Global MMLU: Understanding and Addressing Cultural and Linguistic Biases in Multilingual Evaluation*. arXiv preprint arXiv:2412.03304.

7. Alexandra Institute (2025). *M_ARC*. https://huggingface.co/datasets/alexandrainst/m_arc

8. Ponti, E. M., Glavaš, G., Mrkšić, N., & Korhonen, A. (2020). XCOPA: A multilingual dataset for causal commonsense reasoning. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2362–2376. https://doi.org/10.18653/v1/2020.emnlp-main.185

9. Lin, B. Y., Lee, S., Qiu, M., & Ren, X. (2021). Common sense beyond English: Evaluating and improving multilingual language models for commonsense reasoning. *Proceedings of the 16th Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 3820–3833. https://doi.org/10.18653/v1/2021.naacl-main.298

10. Claws Lab. (2025). *XLingHealth* [Data set]. Hugging Face. https://huggingface.co/datasets/claws-lab/XLingHealth

11. Briakou, Eleftheria, Colin Cherry, and George Foster. "Searching for needles in a haystack: On the role of incidental bilingualism in PaLM's translation capability." *arXiv preprint arXiv:2305.10266* (2023).

12. Chen, Pinzhen, et al. "Is it good data for multilingual instruction tuning or just bad multilingual evaluation for large language models?." *arXiv preprint arXiv:2406.12822* (2024).

13. Hu, J., Ruder, S., Siddhant, A., Neubig, G., Firat, O., & Johnson, M. (2020). *XTREME: A massively multilingual multi-task benchmark for evaluating cross-lingual generalization*. In Proceedings of ICML (or MLR Press).

14. Agrawal, A. S., Fazili, B., & Jyothi, P. (2024). *Translation errors significantly impact low-resource languages in cross-lingual learning*. In Y. Graham & M. Purver (Eds.), Proceedings of the 18th Conference of the European Chapter of the ACL (Vol. 2, Short Papers) (pp. 319–329). Association for Computational Linguistics.

15. Li, Z., Shi, Y., Liu, Z., Yang, F., Payani, A., Liu, N., & Du, M. (2025). *Language Ranker: A metric for quantifying LLM performance across high and low-resource languages*. AAAI (preprint).

16. Chang, T. A., Arnett, C., Tu, Z., & Bergen, B. K. (2024). *When is multilinguality a curse? Language modeling for 250 high- and low-resource languages*. In Proceedings of EMNLP. arXiv preprint.

Appendix

A. Baseline vs English COT evaluation of XCOPA

Language	XCOPA	X-CSQA	X-CODAH
eu	0.4725	0.212	0.28
ja	0.6075	0.34	0.276667
pt	0.705	0.37	0.31
it	0.6525	0.342	0.326667
ru	0.6925	0.354	0.316667
de	0.71	0.374	0.326667
gn	0.4875	0.212	0.276667
lb	0.5275	0.223	0.266667
ba	0.5025	0.204	0.26
mai	0.535	0.242	0.246667
kk	0.53	0.199	0.28
lo	0.465	0.181	0.22
oc	0.61	0.286	0.303333
sd	0.4975	0.211	0.263333
tt	0.48	0.195	0.24
ug	0.505	0.194	0.266667
my	0.4875	0.211	0.263333
lt	0.485	0.225	0.263333
azb	0.5575	0.204	0.26
si	0.505	0.172	0.26

Language	XCOPA	X-CSQA	X-CODAH
eu	0.5	0.203	0.263333
ja	0.6025	0.376	0.38
pt	0.715	0.421	0.42
it	0.7175	0.421	0.466667
ru	0.6575	0.431	0.443333
de	0.65	0.441	0.493333
gn	0.495	0.214	0.253333
lb	0.5275	0.253	0.32
ba	0.53	0.204	0.316667
mai	0.5175	0.259	0.32
kk	0.5175	0.218	0.28
lo	0.455	0.216	0.25
oc	0.635	0.337	0.323333
sd	0.52	0.218	0.293333
tt	0.575	0.206	0.273333
ug	0.52	0.221	0.293333
my	0.49	0.202	0.28
lt	0.5225	0.256	0.276667
azb	0.475	0.194	0.263333
si	0.49	0.2	0.3

Table 7: Baseline vs English COT evaluation of XCOPA
(Languages: de, ja, pt, it, ru, eu, gn, lb, ba, mai, kk, lo, oc, sd, tt, ug, my, lt, azb, si)

B. Token Count Statistic

Language	token_count	token_pct	accuracy	rank_token	rank_accuracy
en	178,269,082.00	0.952238	0.571	1	1
fr	1,406,111.00	0.007511	0.392	2	3
de	1,302,288.00	0.006956	0.374	3	4
es	1,072,049.00	0.005726	0.406	4	2
it	703,479.00	0.003758	0.342	5	9
ru	526,991.00	0.002815	0.354	6	8
pt	382,306.00	0.002042	0.37	7	6
nl	226,390.00	0.001209	0.341	8	10
ca	206,862.00	0.001105	0.311	9	17
pl	192,912.00	0.00103	0.286	10	26
ceb	192,396.00	0.001028	0.252	11	40
ja	192,224.00	0.001027	0.34	12	11
zh	184,040.00	0.000983	0.336	13	12
tr	183,908.00	0.000982	0.29	14	25
sv	158,609.00	0.000847	0.294	15	22

Language	token_count	token_pct	accuracy	rank_token	rank_accuracy
en	729,015,568.00	0.679844	0.571	1	1
ru	67,797,191.00	0.063224	0.354	2	8
es	30,959,373.00	0.028871	0.406	3	2
zh	29,807,725.00	0.027797	0.336	4	12
fr	27,377,465.00	0.025531	0.392	5	3
de	21,144,342.00	0.019718	0.374	6	4
it	19,934,681.00	0.01859	0.342	7	9
tr	13,487,453.00	0.012578	0.29	8	25
pt	12,171,325.00	0.01135	0.37	9	6
ja	11,038,205.00	0.010294	0.34	10	11
ko	9,155,948.00	0.008538	0.305	11	19
uk	7,983,709.00	0.007445	0.292	12	24
ceb	7,384,591.00	0.006887	0.252	13	40
vi	6,274,382.00	0.005851	0.312	14	16
ta	5,984,867.00	0.005581	0.212	15	63

Table 8: Token count vs Accuracy for XCSQA dataset
(Pre-training vs Post-training)

language	token_count	token_pct	accuracy	rank_token	rank_accuracy
en	178,269,082	0.941913	0.613	1	1
fr	1,406,111	0.007429	0.252	2	4
de	1,302,288	0.006881	0.256	3	3
es	1,072,049	0.005664	0.26	4	2
it	703,479	0.003717	0.25	5	5
ru	526,991	0.002784	0.207	6	32
pt	382,306	0.00202	0.247	7	6
nl	226,390	0.001196	0.244	8	7
ca	206,862	0.001093	0.2405	9	8
ceb	192,396	0.001017	0.2385	10	10
pl	192,912	0.001019	0.24	11	9
ja	192,224	0.001016	0.204	12	39
zh	184,040	0.000972	0.206	13	34
tr	183,908	0.000972	0.238	14	11
sv	158,609	0.000838	0.238	15	11

language	token_count	token_pct	accuracy	rank_token	rank_accuracy
en	729,015,568	0.679844	0.613	1	1
ru	67,797,191	0.063224	0.207	2	32
es	30,959,373	0.028871	0.26	3	2
zh	29,807,725	0.027797	0.206	4	34
fr	27,377,465	0.025531	0.252	5	4
de	21,144,342	0.019718	0.256	6	3
it	19,934,681	0.01859	0.25	7	5
tr	13,487,453	0.012578	0.238	8	11
pt	12,171,325	0.01135	0.247	9	6
ja	11,038,205	0.010294	0.204	10	39
ko	9,155,948	0.008538	0.184	11	55
uk	7,983,709	0.007445	0.208	12	26
ceb	7,384,591	0.006887	0.2385	13	10
vi	6,274,382	0.005851	0.2245	14	18
ta	5,984,867	0.005581	0.195	15	43

Table 9: Token count vs Accuracy (Pre-training & Post-Training) - MMLU for 100 languages

language	token_count	token_pct	accuracy	rank_token	rank_accuracy
en	178,269,082	0.948576	0.6314	1	1
fr	1,406,111	0.007482	0.2945	2	9
de	1,302,288	0.00693	0.3129	3	6
es	1,072,049	0.005704	0.3075	4	7
it	703,479	0.003743	0.33	5	4
ru	526,991	0.002804	0.2341	6	36
pt	382,306	0.002034	0.3366	7	3
nl	226,390	0.001205	0.2714	8	15
ca	206,862	0.001101	0.2861	9	13
pl	192,912	0.001026	0.2879	10	12
ceb	192,396	0.001024	0.2532	11	23
ja	192,224	0.001023	0.2117	12	62
zh	184,040	0.000979	0.2338	13	37
tr	183,908	0.000979	0.2749	14	14
sv	158,609	0.000844	0.3377	15	2

language	token_count	token_pct	accuracy	rank_token	rank_accuracy
en	729,015,568	0.662988	0.6314	1	1
ru	67,797,191	0.061657	0.2341	2	36
es	30,959,373	0.028155	0.3075	3	7
zh	29,807,725	0.027108	0.2338	4	37
fr	27,377,465	0.024898	0.2945	5	9
de	21,144,342	0.019229	0.3129	6	6
it	19,934,681	0.018129	0.33	7	4
tr	13,487,453	0.012266	0.2749	8	14
pt	12,171,325	0.011069	0.3366	9	3
ja	11,038,205	0.010039	0.2117	10	62
ko	9,155,948	0.008327	0.1812	11	93
uk	7,983,709	0.007261	0.2247	12	45
ceb	7,384,591	0.006716	0.2532	13	23
ar	6,886,664	0.006263	0.2194	14	49
fa	6,790,399	0.006175	0.2117	15	62

Table 10: Token count vs Accuracy (Pre-training & Post-Training) - M_ARC for 100 languages

Language	Pre-train Token Count	Pre-train Token %	BERTScore	Token Rank
en	95000000	14.329	0.98	1
fr	65000000	9.804	0.97	5
es	70000000	10.558	0.98	3
de	60000000	9.05	0.977	6
it	58000000	8.748	0.96	7
ru	68000000	10.256	0.973	4
zh	72000000	10.86	0.98	2
pt	40000000	6.033	0.945	8
nl	35000000	5.279	0.929	9
pl	30000000	4.525	0.93	10
tr	28000000	4.223	0.92	11
hi	20000000	3.017	0.909	12
sw	12000000	1.81	0.905	13
ta	8000000	1.207	0.882	14
ceb	2000000	0.302	0.876	15

Language	Post-train Token Count	Post-train Token %	BERTScore	Token Rank
en	96900000	14.152	0.99	1
fr	66950000	9.778	0.99	5
es	72100000	10.53	0.99	3
de	61499999	8.982	0.984	6
it	58985999	8.615	0.984	7
ru	70040000	10.229	0.99	4
zh	74880000	10.936	0.99	2
pt	42000000	6.134	0.957	8
nl	37100000	5.418	0.94	9
pl	30989999	4.526	0.944	10
tr	28979999	4.232	0.948	11
hi	21500000	3.14	0.923	12
sw	12480000	1.823	0.921	13
ta	8199999	1.198	0.906	14
ceb	2100000	0.307	0.894	15

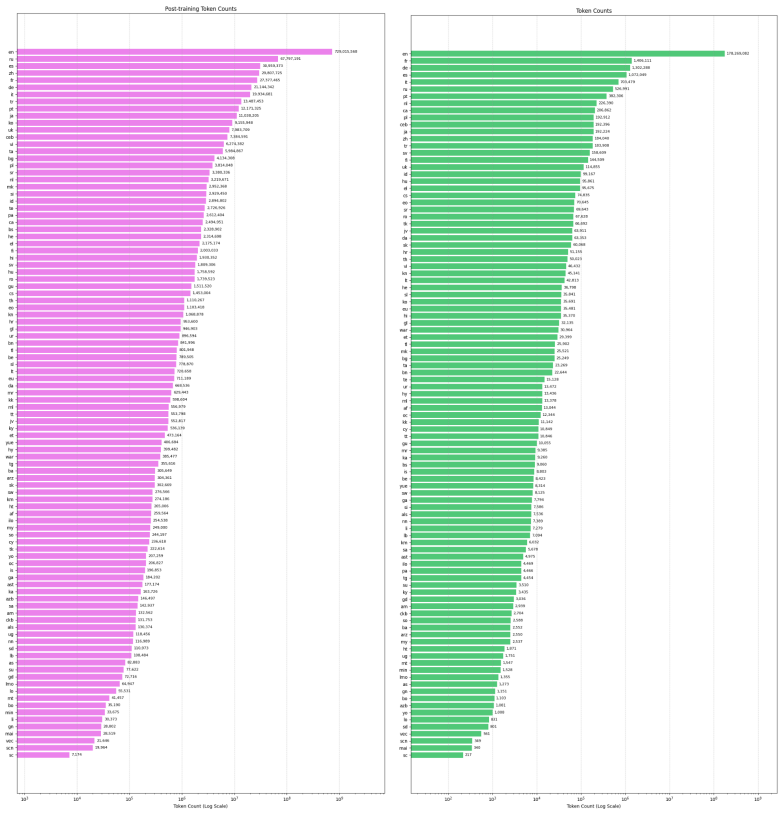
Table 11: Token Count vs BERTScore (Pre and Post training) - HealthcareQA for 100 languages

Token Count analysis of Math Dataset at script level

	script	math avg score	token_count
0	Latn	0.159375	185560665.0
1	Cyrl	0.122727	803111.0
2	Jpan	0.116500	192224.0
3	Hans	0.266500	184040.0
4	Grek	0.148000	95675.0
5	Deva	0.085500	50773.0
6	Thai	0.157000	50023.0
7	Knda	0.108500	45141.0
8	Hebr	0.142250	40179.0
9	Hang	0.173000	35691.0
10	Beng	0.053500	23917.0
11	Taml	0.129500	23269.0
12	Arab	0.110500	22359.0
13	Telu	0.148500	15128.0
14	Armn	0.078000	13436.0
15	Mlym	0.104000	13378.0
16	Gujr	0.143000	10055.0
17	Geor	0.067500	9260.0
18	Hant	0.165500	8314.0
19	Sinh	0.089000	7586.0
20	Khmr	0.030500	6032.0
21	Guru	0.149000	4466.0
22	Ethi	0.063500	2939.0
23	Mymr	0.015000	2537.0
24	Tibt	0.014000	1103.0
25	Laoo	0.051500	831.0

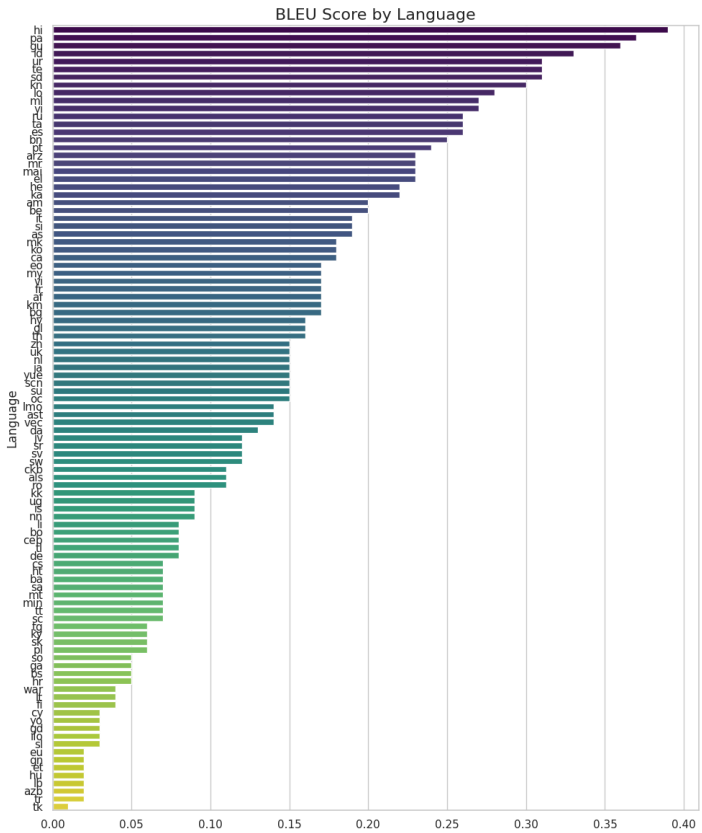
Rank correlation for 25 scripts: 0.559

C. Language-wise Token Count distribution in sampled Pretraining and Post training dataset



D. Back translation results - NLLB

NLLB Translation accuracy on Math Datasets via back translation for 100 languages



MGSM

- Average BLEU: 0.1405
- Average METEOR: 0.3202
- Average ROUGE-L: 0.4155
- Average TER: 74.9228

MSVAMP

- Average BLEU: 0.1029
- Average METEOR: 0.3193
- Average ROUGE-L: 0.4159
- Average TER: 73.8598

E. Back translation results - Hunyuan 7B

LANGUAGE	BLEU	CHRPFF	COSINE	ROUGE_L
bn	0.256181	0.574868	0.809109	0.619212
bo	0.266522	0.509053	0.731337	0.563006
es	0.390569	0.666003	0.852064	0.700342
hi	0.300708	0.606968	0.833798	0.646063
kk	0.204965	0.475504	0.685534	0.520347
km	0.189731	0.472899	0.722797	0.528163
ko	0.327107	0.634561	0.861739	0.683873
mn	0.135339	0.382553	0.607053	0.446396
mr	0.274759	0.574547	0.820632	0.627931
ug	0.248294	0.527471	0.758506	0.572389
AVERAGE	0.2594175	0.5424427	0.7682569	0.5907722

Table 12: Huyuan 7B back translation (XCOPA)